5 A circuit is set up as shown in Fig. 5.1. The circuit contains a battery of e.m.f. 12 V, and a fixed resistor r of resistance 6 Ω and length L_r = 0.50 m. A movable metal rod M, of length L_M = 0.60 m and resistance 12 Ω, is placed on a conducting track at a distance d = 4.0 cm from r. The conducting track is assumed to be smooth and has zero resistance.

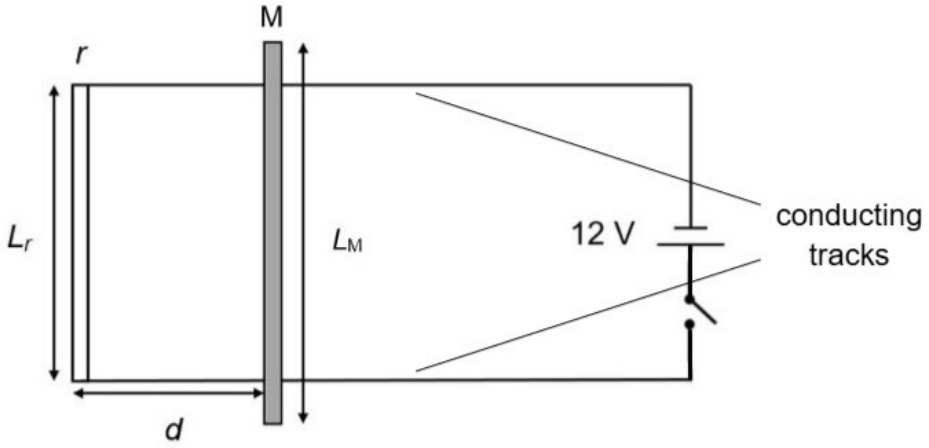


Fig. 5.1 (not to scale)

The switch is now closed.

(a) Show that the magnetic flux density B at the metal rod M due to the current in resistor r is 10 μT.

[1]

(b) Calculate the magnetic force experienced by M and state its direction.

magnetic force = ... N

direction: ..... [3]

(c) M in **(b)** is held stationary and a magnetic field of flux density 1.0 mT is now directed out of page throughout the circuit in Fig. 5.1. M is now released.

Describe and explain the initial motion of M.

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..... [2]

(d) A proton enters the magnetic field *B* at an angle to the field of less than 90°.

The proton follows the path shown in Fig. 5.2.

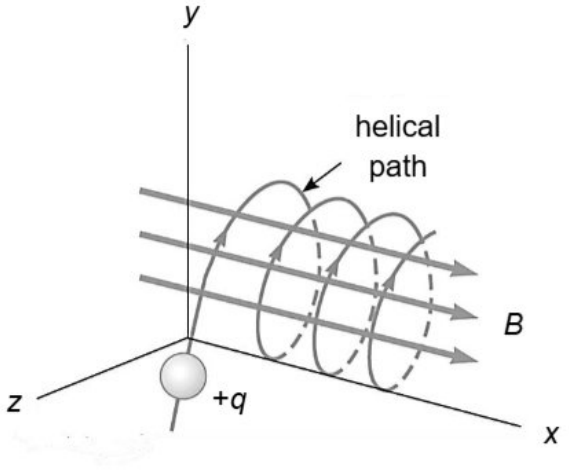


Fig. 5.2

(i) Explain why the proton follows this helical path.

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..... [2]

(ii) Fig. 5.3 shows a magnetic bottle, where a non-uniform magnetic field is produced between two current carrying coils. A proton is initially directed perpendicularly into the page with speed v at the position shown.

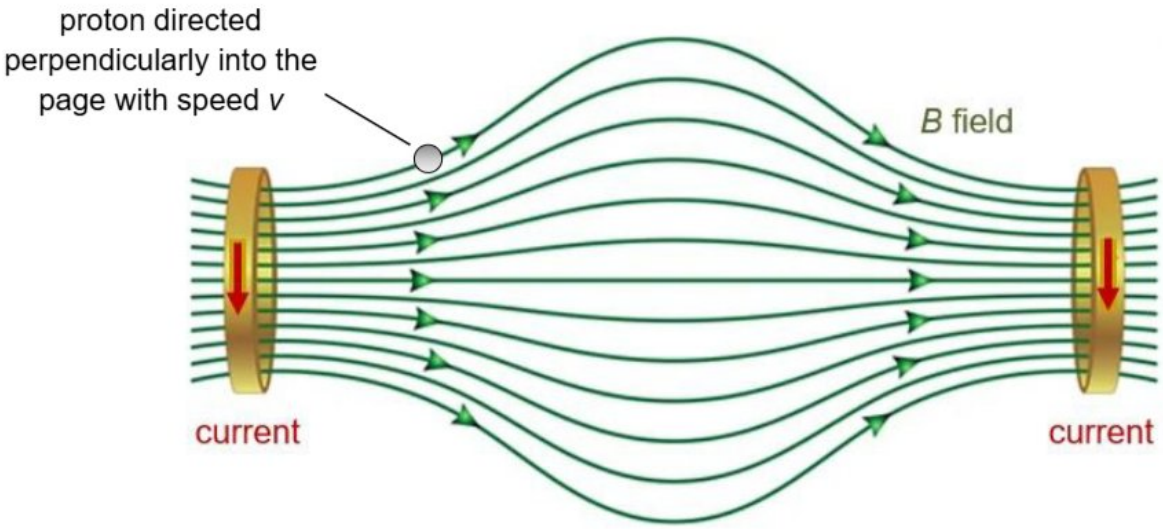


Fig. 5.3

Describe and explain the subsequent motion of the proton in the magnetic bottle, assuming the proton remains within the field region throughout.

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..... [2]

[Total: 10]